Register No:												

DEMO INSTITUTE OF SCIENCE AND TECHNOLOGY

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Demo City, Tech State - 123 456, India

MODEL EXAMINATION NOVEMBER 2026
M.Sc MATHEMATICS I SEMESTER
GRAPH THEORY
SET-I

Date: 17.10.2026 FN	Time: 3:00 HOURS	Maximum Marks: 75

PART-A (15 x 1 = 15 Marks)
Answer all questions

1	What is the primary focus of directed graphs as outlined in the basic concepts? **Answer Key:** Directed edges				**CO1**	**L1**
	A	Edge weights	**C**	Unbounded cycles		
	B	Directed edges	**D**	Planar faces		
2	What term is fundamentally used in basic graph theory to describe the process of creating a new graph by combining the vertex sets and edge sets of two existing graphs? **Answer Key:** Graph Operation				**CO1**	**L1**
	A	Graph Traversal	**C**	Graph Coloring		
	B	Graph Operation	**D**	Graph Planarity		
3	What characterizes a tournament in directed graphs? **Answer Key:** Every pair of distinct vertices is joined by a directed edge				**CO1**	**L1**
	A	Every pair of distinct vertices is joined by a directed edge	**C**	All vertices have equal degrees		
	B	No edges are directed	**D**	It is always planar		
4	What does vertex connectivity measure in a graph? **Answer Key:** Minimum vertices to remove to disconnect				**CO2**	**L1**
	A	Number of edges to disconnect	**C**	Total paths available		
	B	Minimum vertices to remove to disconnect	**D**	Length of the longest cycle		
5	Who formulated the famous formula for counting labeled trees on n vertices? **Answer Key:** Cayley				**CO2**	**L1**
	A	Euler	**C**	Kuratowski		
	B	Cayley	**D**	Heawood		

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6	What is a defining characteristic of trees in graph theory? **Answer Key:** They are connected acyclic graphs				**CO2**	**L1**
	A	They contain cycles	**C**	They have no vertices		
	B	They are connected acyclic graphs	**D**	They are always regular		
7	What distinguishes an Eulerian graph? **Answer Key:** Contains a closed trail visiting every edge				**CO3**	**L1**
	A	Contains a closed trail visiting every edge	**C**	Is strictly planar		
	B	Contains a Hamiltonian cycle	**D**	Has no independent sets		
8	What is a Hamiltonian graph characterized by? **Answer Key:** A cycle visiting every vertex				**CO3**	**L1**
	A	A cycle visiting every vertex	**C**	A tree structure		
	B	A path visiting every edge	**D**	An empty edge set		
9	What do matchings and factors deal with in graph theory? **Answer Key:** Independent edge subsets				**CO3**	**L1**
	A	Vertex colorings	**C**	Planar embeddings		
	B	Independent edge subsets	**D**	Chromatic polynomials		
10	If a graph is triangle-free, what can be applied regarding its vertex coloring properties? **Answer Key:** Critical graph reductions				**CO4**	**L1**
	A	Maximum clique bounds	**C**	Eulerian trail tests		
	B	Critical graph reductions	**D**	Cayley tree formulas		
11	What does the chromatic polynomial $P(G, k)$ of a graph G primarily calculate? **Answer Key:** The number of ways to color the graph using k colors				**CO4**	**L1**
	A	The total number of edges in the graph	**C**	The minimum number of colors required for the graph		
	B	The number of ways to color the graph using k colors	**D**	The maximum degree of a vertex in the graph		
12	What defines a critical graph in vertex coloring? **Answer Key:** Its chromatic number drops if any vertex is removed				**CO4**	**L1**
	A	Its chromatic number drops if any vertex is removed	**C**	It is nonplanar		
	B	It has no edges	**D**	It contains K5		

13	Which graphs are famously known as nonplanar base graphs according to Kuratowski's theorem? **Answer Key:** K5 and K3,3				**CO5**	**L1**
	A	K5 and K3,3	**C**	K1,1 and K2,2		
	B	K4 and K2,3	**D**	K3 and K4		
14	What does Euler's Formula relate in a connected planar graph? **Answer Key:** Vertices, edges, and faces				**CO5**	**L1**
	A	Vertices, edges, and faces	**C**	Edges and vertex connectivity		
	B	Coloring and chromatic number	**D**	Spanning trees and Cayley numbers		
15	Which theorem bounds the chromatic number of planar graphs to five? **Answer Key:** Heawood Five-Colour Theorem				**CO5**	**L1**
	A	Heawood Five-Colour Theorem	**C**	Kuratowski's Theorem		
	B	Four-Colour Theorem	**D**	Cayley's Formula		

PART-B (2 x 5 = 10 Marks)
Answer any TWO questions

16	Illustrate the fundamental differences between directed graphs and tournament graphs with examples. **Answer Key:** 1. Directed graphs allow cycles and multiple edges. 2. Tournaments are directed complete graphs. 3. Every pair has exactly one directed edge.	**CO1**	**L?**
17	Calculate the vertex and edge connectivity for a given cycle graph C_n. **Answer Key:** 1. Vertex connectivity of C_n is 2.2. Edge connectivity of C_n is 2.3. Removal of 2 vertices or edges disconnects the cycle.	**CO2**	**L3**
18	Compare Eulerian and Hamiltonian properties regarding edge traversal and vertex visitation constraints. **Answer Key:** Eulerian focuses on edge covering; Hamiltonian focuses on vertex sequencing; NP-complete complexity difference.	**CO3**	**L2**
19	Apply the vertex colouring principle to find the chromatic number of a triangle-free graph. **Answer Key:** Use independent sets to assign minimum colours without adjacent conflicts.	**CO4**	**L3**
20	State Euler's formula for connected planar graphs. **Answer Key:** n - e + f = 2, where n is vertices, e is edges, f is faces.	**CO5**	**L1**

PART-C (5 x 10 = 50 Marks)
Answer all questions

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21	**A**	Analyze the structural differences between general directed graphs and tournaments, focusing on path properties and connectedness. **Answer Key:** Key value points: Directed edges, asymmetry in tournaments, Hamiltonian paths in tournaments, strong connectedness criteria.	**CO1**	**L4**
	OR			
	B	Evaluate how basic graph operations alter the degree sequences and connectedness of subgraphs in complex networks. **Answer Key:** Key value points: Union, intersection, ring sum, impact on vertex degrees, preservation of connectedness.	**CO1**	**L5**
22	**A**	Analyze the relationship between vertex cuts, edge cuts, and the overall connectivity metrics of a graph. **Answer Key:** Key value points: Minimum vertex cuts, edge connectivity bounds, Menger's theorem implications, block structures.	**CO2**	**L4**
	OR			
	B	Formulate a proof using Cayley's formula to count the spanning trees of labeled graphs with specific degree constraints. **Answer Key:** Key value points: Prüfer codes, labeled trees, tree counting bijection, algebraic enumeration.	**CO2**	**L6**
23	**A**	Examine the necessary and sufficient conditions distinguishing Eulerian graphs from Hamiltonian graphs in undirected networks. **Answer Key:** Key value points: Degree parity for Eulerian, Dirac's and Ore's theorems for Hamiltonian, traversal differences.	**CO3**	**L4**
	OR			
	B	Assess the impact of edge and vertex independent sets on maximum matchings and factors within bipartite graphs. **Answer Key:** Key value points: Hall's Marriage Theorem, Tutte's 1-factor theorem, maximal independent sets, vertex covers.	**CO3**	**L5**
24	**A**	Analyze critical graphs and triangle-free graphs to deduce their chromatic number properties and limitations. **Answer Key:** Key value points: Edge-critical graphs, Mycielski's construction, independence number bounds, triangle-free coloring.	**CO4**	**L4**
	OR			
	B	Design a chromatic polynomial for a given graph and evaluate its coefficients to interpret coloring solutions. **Answer Key:** Key value points: Deletion-contraction recurrence, polynomial expansion, coefficient alternating signs, evaluating proper colorings.	**CO4**	**L6**

25	**A**	Examine Euler's formula and its consequences to prove the nonplanarity of complete and bipartite graphs. **Answer Key:** Key value points: Planar embeddings, face boundaries, K_5 and $K_{3,3}$ edge-vertex inequalities, bridge to Kuratowski's theorem.	**CO5**	**L4**
	OR			
	B	Formulate a comparative critique showing how Heawood's five-colour approach contrasts with the computer-assisted resolution of the four-colour theorem. **Answer Key:** 1. Contrasts traditional topological proofs with brute-force computer verification. 2. Highlights conceptual elegance vs exhaustive case checking. 3. Assesses the reliability of algorithmic map coloring.	**CO5**	**L6**

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